

AgroPulse-EO — Business Model Canvas

Annex – Initial Exploitation Pathway / Business Model Canvas – ABIL EXPLORE Kenya Open Call

Customer Segments	• Primary: individual Kenyan smallholder farmers and farmer 'chama' (savings/social) groups already using AgroPulse for farm management, drone scouting and AI crop diagnosis. • Secondary / near-term roadmap: greenhouse and commercial horticulture operators (higher willingness/ability to pay for precision monitoring). • Institutional: county agricultural extension offices and agri-finance/insurance providers, who consume aggregate, anonymised farm-risk signals rather than raw farmer data.
Value Propositions	• On-demand, AI-powered crop diagnosis (disease/pest/nutrient detection with treatment recommendations) from a drone or phone photo – already live and in use. • Free, continuous satellite-based early-warning triage between physical drone visits, calibrated against AgroPulse's own real, local ground-truth data rather than generic global thresholds (the capability this ABIL EXPLORE activity will build and validate). • Farm management and a trusted social/savings ('chama') layer in one platform, rather than separate disconnected tools. • A credible, near-zero-marginal-cost path to extend coverage to greenhouse/commercial horticulture using the same validated data-fusion approach.
Channels	• Direct distribution via the AgroPulse Android mobile app. • Farmer 'chama' / savings-group networks (existing trust-based distribution channel). • County agricultural extension offices (potential institutional distribution partner). • Farmer-to-farmer word-of-mouth within the existing active user base.
Customer Relationships	• Self-service via the mobile app for individual farmers. • Community-mediated trust and adoption through chama groups. • Direct institutional partnership/licensing relationships for county offices and agri-finance providers (once validated).
Revenue Streams	• Currently pre-revenue and self-funded by the Founder. • Planned: modest farmer/chama subscription or per-diagnosis fee, in line with what this user base already pays for other digital services. • Planned: institutional licensing of aggregate, anonymised triage/risk signals to county extension offices and agri-finance/insurance providers. • Planned: a higher-ARPU subscription for greenhouse/commercial horticulture precision monitoring, once that layer is activated.
Key Resources	• The existing, operating AgroPulse platform: FastAPI/Python backend and Flutter mobile app. • A real, geotagged farm-boundary and AI-diagnosis ground-truth dataset built from active Kenyan smallholder use. • A live IoT sensor-ingestion backend (device registration, environmental logging, alerting). • The Founder's full-stack technical capability, demonstrated by building and operating the platform solo to date. • Open-access satellite EO data (Copernicus / Google Earth Engine).
Key Activities	• Backend and mobile software development and maintenance. • AI-powered crop diagnosis (multi-provider LLM integration). • Drone-based farm scouting, boundary mapping and photo capture. • Satellite EO data ingestion and calibration against real ground truth (this activity's core technical work). • Farmer/chama community engagement and field validation.

Key Partners	• ABIL / ESA / Kenya Space Agency – EO ecosystem access, technical mentoring and business coaching. • A contracted GIS/remote-sensing specialist (to be engaged under this activity). • Farmer 'chama' groups – trust network and distribution channel. • Copernicus / Google Earth Engine – open EO data providers. • Roadmap: county agricultural extension offices, agri-finance/insurance providers.
Cost Structure	• Personnel: the Founder plus contracted specialists (e.g. the GIS/remote-sensing role funded under this activity). • Cloud/compute hosting for the backend, AI inference and EO data processing. • AI/LLM API usage costs (per-diagnosis inference). • Field validation and in-Kenya travel. • Currently self-funded; ABIL EXPLORE funding is requested specifically for the satellite-EO validation activity described in the Technical and Activity Proposals.

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